

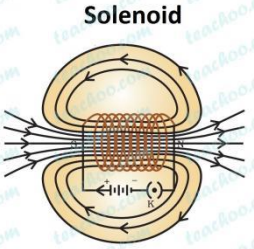
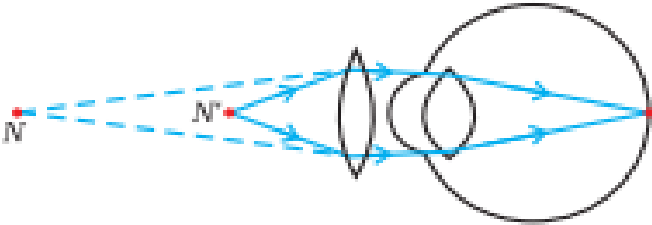
10PB24SCI03 MS
KENDRIYA VIDYALAYA SANGATHAN ,ERNAKULAM REGION
PRE BOARD EXAMINATION (2023-24)
CLASS-X -SCIENCE (086) MARKING SCHEME

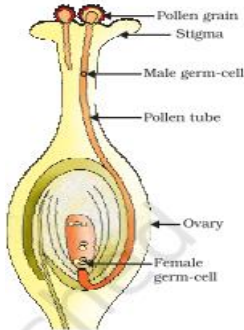
MAX.M:80

TIME:3Hrs

Section-A		
Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions		
1	C.Decomposition of lead nitrate to produce nitrogen dioxide	1
2	B.3,4,4	1
3	C.(a) and (c)	1
4	B.(a) and (c)	1
5	C.absorb moisture from the gas	1
6	A. $B > A > C > D$	1
7	B.copper and tin	1
8	D.B	1
9	B. contraction of left ventricle	1
10	B.small intestine	1
11	C.cerebellum	1
12	B.Round Yellow	1
13	D.I,II and IV	1
14	B. contracts and lens becomes thicker	1
15	C.secondary consumer to tertiary consumer	1
16	C.3J	1
17	C.A is true but R is false	1
18	A. Both A and R are true, and R is the correct explanation of A.	1
19	C.A is true but R is false.	1
20	D. A is false but R is true	1
Section-B		
Question No. 21 to 26 are very short answer questions		
21	a)Substance oxidized : HCl and Substance reduced : MnO_2 b) $Al_2(SO_4)_3 + 3BaCl_2 \rightarrow 2AlCl_3 + 3BaSO_4$	$\frac{1}{2} + \frac{1}{2}$ 1
22	1. The exit of food from the stomach is regulated by a sphincter muscle which releases it in small amounts into the small intestine. 2. The exit of waste material is regulated by the anal sphincter. OR (i) Absorption of light energy by chlorophyll. (ii) Conversion of light energy to chemical energy and (iii) splitting of water molecules into hydrogen and oxygen. (iv) Reduction of carbon dioxide to carbohydrates	1+1 $\frac{1}{2} \times 4$
23	The embryo gets nutrition from the mother's blood with the help of a special tissue called placenta. This is a disc which is embedded in the uterine wall. It contains villi on the embryo's side of the tissue. On the mother's side are blood spaces, which surround the villi. This provides a large surface area for glucose and oxygen to pass from the mother to the embryo. The developing embryo will also generate waste substances which can be removed by transferring them into the mother's blood through the placenta.	1 1
24	a) focal length is +0.2m /+20cm and lens is convex b) (i)at 15 cm ,object is kept between F and O (ii)at 20cm object is at F or at 22cm object is between 2F and F	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

25	a) The law implies that heat produced in a resistor is (i) directly proportional to the square of current for a given resistance, (ii) directly proportional to resistance for a given current, and (iii) directly proportional to the time for which the current flows through the resistor. b) Any two applications OR a) Any two advantages. b) In series connection current through all components of the circuit is constant, Hence an ammeter connected in series can measure the correct magnitude of current flowing through the circuit.	1 $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ 1
26	a) Ozone (O_3) is a molecule formed by three atoms of oxygen. Ozone at the higher levels of the atmosphere is a product of UV radiation acting on oxygen (O_2) molecule. The higher energy UV radiations split apart some molecular oxygen (O_2) into free oxygen (O) atoms. These atoms then combine with the molecular oxygen to form ozone or Relevant equations B) At the higher levels of the atmosphere, ozone performs an essential function. It shields the surface of the earth from ultraviolet (UV) radiation from the Sun.	$\frac{1}{2} + \frac{1}{2}$ 1
Section-C Question No. 27 to 33 are short answer questions		
27	a) calcium sulphate hemi hydrate b) On heating gypsum at 373 K, it loses water molecules and becomes calcium sulphate hemihydrate . <i>Gypsum</i> $CaSO_4 \cdot 2H_2O \xrightarrow{373K} CaSO_4 \cdot \frac{1}{2}H_2O + \frac{3}{2}H_2O$ <i>POP</i> c) Plaster of Paris (POP) should be stored in a moisture-proof container because Plaster of Paris, a powdery mass, absorbs water (moisture) to form a hard solid known as gypsum.	$\frac{1}{2}$ $\frac{1}{2}$ 1 1
28	Electrolytic Refining: In this process, the impure metal is made the anode and a thin strip of pure metal is made the cathode. On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte. An equivalent amount of pure metal from the electrolyte is deposited on the cathode. The soluble impurities go into the solution, whereas, the insoluble impurities settle down at the bottom of the anode and are known as anode mud. Figure with labelling OR a) Some metal oxides, such as aluminium oxide, zinc oxide show both acidic as well as basic behaviour. Such metal oxides which react with both acids as well as bases to produce salt and water are known as amphoteric oxides. b) Correct equations for the reaction of aluminium oxide or zinc oxide with an acid and a base	1 2 1 1+1
29	Figure 5.8 Break-down of glucose by various pathways	1 1 1
30	a) Male individual has 46 chromosomes but because the gametes produced after meiosis have half the no. of chromosomes, It is the fusion of this gametes which leads to an offspring with 46	1

	chromosomes. b) Correct flow chart depicting the formation of a girl child.	2
31	 a) b) The field lines inside the solenoid are in the form of parallel straight lines. This indicates that the magnetic field is the same at all points inside the solenoid. That is, the field is uniform inside the solenoid	2 1
32	a) Hypermetropia is also known as far-sightedness. A person with hypermetropia can see distant objects clearly but cannot see nearby objects distinctly. b) This defect arises either because (i) the focal length of the eye lens is too long, or (ii) the eyeball has become too small. c)  (c) Correction for Hypermetropic eye	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ 1
33	Correct circuit diagram Ammeter reading $I = V/R = 6/25 = 0.24A$ Voltmeter reading $V = IR = 0.24 \times 12 = 2.88V$	2 $\frac{1}{2}$ $\frac{1}{2}$
Section-D Question No. 34 to 36 are long answer questions.		
34	(a) A is ethanoic acid and B is ester (b) $CH_3COOH + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + H_2O$ (c) perfumes, flavouring agents (d) sodium acetate, carbon dioxide and water is formed $2CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + CO_2 + H_2O$ (e) ethanoic acid reacts with a base such as sodium hydroxide to give a salt (sodium ethanoate or commonly called sodium acetate) and water and correct equation OR a) The ionic-end of soap interacts with water while the carbon chain interacts with oil. The soap molecules, thus form structures called micelles. b) Diagram showing formation of micelles. c) Most dirt is oily in nature and oil does not dissolve in water. The molecules of soap are sodium or potassium salts of long-chain carboxylic acids. The ionic-end of soap interacts with water while the carbon chain interacts with oil. The soap molecules, thus form structures called micelles where one end of the molecules is towards the oil droplet while the ionic-end faces outside. This forms an emulsion in water. The soap micelle thus helps in pulling out the dirt in water and we can wash our clothes clean. d) On treating with sodium hydroxide, which is an alkali, the ester is converted back to alcohol and sodium salt of carboxylic acid. This reaction is known as saponification because it is used in the preparation of soap. Correct equation for the reaction	$\frac{1}{2} + \frac{1}{2}$ 1 $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ 1 1 2 $\frac{1}{2} + \frac{1}{2}$

35	a) A-prostate gland B-seminal vesicle They add secretions so that the sperms are in a fluid which makes their transport easier and this fluid also provides nutrition. b) Any three contraceptive methods with explanation OR a) After the pollen lands on a suitable stigma, it has to reach the female germ cells which are in the ovary. For this, a tube grows out of the pollen grain and travels through the style to reach the ovary. b) Diagram with labelling  Figure 7.8 Germination of pollen on stigma	$\frac{1}{2}+\frac{1}{2}$ $\frac{1}{2}+\frac{1}{2}$ 1x3 2 3
36	a) Ohm's law states that the electric current flowing through a conductor is directly proportional to the potential difference across its two ends, when other physical conditions like temperature remains constant. b) Activity to verify ohms law NCERT text book c) Potential difference $V = 60\text{ V}$, current $I = 4\text{ A}$. According to Ohm's law, $R = V/I = 60/4 = 15\Omega$ When the potential difference is increased to 120 V the current is given by $I = V/R = 120/15 = 8\text{ A}$. d) $1/R = 1/R/5 + 1/R/5 + 1/R/5 + 1/R/5 + 1/R/5$ $= 25/R$ $R' = R/25 = 25$ OR a) It is the property of a conductor to resist the flow of charges through it. b) Any two factors affecting resistance c) Resistance of a uniform metallic conductor is directly proportional to its length (l) and inversely proportional to the area of cross-section (A). That is, $R \propto l$ $R \propto 1/A$ we get $R \propto l/A$ or, $R = \rho l/A$ where ρ (rho) is a constant of proportionality and is called the electrical resistivity of the material of the conductor. $\rho = RA/l$ c) $r = 0.5/2 \times 10^{-3}$ $\rho = 1.6 \times 10^{-8} \Omega\text{m}$, $R = 10\Omega$ $R = \rho l/A$ $l = R \pi r^2 / \rho$ $= 10 \times 3.14 / 1.6 \times 10^{-8} \times (0.5/2 \times 10^{-3})^2 = 122.6\text{m}$	1 2 $\frac{1}{2}+\frac{1}{2}$ 1 1 $\frac{1}{2}+\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

Section – E

Question No. 37 to 39 are case-based/data -based questions.

